

# Method Metrics

## References

- [jqassistant](#)
- [Neo4j Python Driver](#)

## Effective Method Line Count

### Table 1a - Effective method line count distribution

This table shows the distribution of the effective method line count per artifact. For each artifact the number of methods with effective line count = 1,2,3,... is shown to get an overview of how line counts are distributed over methods.

Only the 15 artifacts with the highest method count and their effective method line count distribution (limited by 40) is shown here. The whole table can be found in the CSV report [Effective\\_Method\\_Line\\_Count\\_Distribution](#).

Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName       | axon-messaging-5.0.2.jar | axon-common-5.0.2.jar | axon-server-connector-5.0.2.jar | axon-eventsourcing-5.0.2.jar | axon-test-5.0.2.jar | axon-modelling-5.0.2.jar | axon-spring-boot-autoconfigure-5.0.2.jar | axon-conversion-5.0.2.jar | axon-update-5.0.2.jar | axon-metrics-micrometer-5.0.2.jar |
|--------------------|--------------------------|-----------------------|---------------------------------|------------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|-----------------------------------|
| effectiveLineCount |                          |                       |                                 |                              |                     |                          |                                          |                           |                       |                                   |
| 1                  | 1935                     | 351                   | 266                             | 305                          | 204                 | 175                      | 181                                      | 101                       | 88                    | 34                                |
| 2                  | 495                      | 141                   | 143                             | 75                           | 69                  | 56                       | 79                                       | 20                        | 10                    | 17                                |
| 3                  | 415                      | 93                    | 77                              | 61                           | 33                  | 30                       | 16                                       | 12                        | 6                     | 15                                |
| 4                  | 189                      | 39                    | 44                              | 53                           | 37                  | 36                       | 6                                        | 8                         | 10                    | 6                                 |
| 5                  | 153                      | 36                    | 33                              | 31                           | 4                   | 23                       | 9                                        | 9                         | 8                     | 2                                 |
| 6                  | 92                       | 36                    | 15                              | 17                           | 14                  | 18                       | 7                                        | 7                         | 8                     | 4                                 |
| 7                  | 67                       | 14                    | 9                               | 15                           | 8                   | 9                        | 5                                        | 5                         | 2                     | 4                                 |
| 8                  | 71                       | 17                    | 15                              | 5                            | 7                   | 9                        | 2                                        | 5                         | 2                     | 5                                 |
| 9                  | 36                       | 11                    | 11                              | 10                           | 7                   | 13                       | 1                                        | 4                         | 4                     | 1                                 |
| 10                 | 29                       | 6                     | 7                               | 7                            | 2                   | 3                        | 1                                        | 1                         | 2                     | 0                                 |
| 11                 | 20                       | 7                     | 2                               | 5                            | 4                   | 2                        | 0                                        | 2                         | 0                     | 0                                 |
| 12                 | 10                       | 7                     | 3                               | 4                            | 7                   | 4                        | 1                                        | 0                         | 1                     | 0                                 |
| 13                 | 8                        | 5                     | 2                               | 4                            | 2                   | 5                        | 1                                        | 1                         | 3                     | 3                                 |
| 14                 | 4                        | 5                     | 2                               | 3                            | 2                   | 1                        | 0                                        | 0                         | 2                     | 0                                 |
| 15                 | 6                        | 1                     | 1                               | 2                            | 3                   | 1                        | 0                                        | 1                         | 0                     | 1                                 |
| 16                 | 7                        | 2                     | 1                               | 1                            | 3                   | 1                        | 0                                        | 2                         | 1                     | 0                                 |
| 17                 | 4                        | 3                     | 2                               | 1                            | 0                   | 2                        | 1                                        | 1                         | 2                     | 0                                 |
| 18                 | 1                        | 1                     | 1                               | 0                            | 2                   | 2                        | 0                                        | 0                         | 1                     | 0                                 |
| 19                 | 3                        | 0                     | 0                               | 1                            | 0                   | 0                        | 0                                        | 1                         | 0                     | 0                                 |
| 20                 | 2                        | 0                     | 1                               | 1                            | 2                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 21                 | 1                        | 0                     | 2                               | 1                            | 1                   | 1                        | 0                                        | 0                         | 2                     | 0                                 |
| 22                 | 3                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 23                 | 1                        | 0                     | 0                               | 0                            | 1                   | 0                        | 1                                        | 0                         | 0                     | 0                                 |
| 24                 | 0                        | 1                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 1                     | 0                                 |
| 25                 | 2                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 26                 | 0                        | 1                     | 0                               | 0                            | 1                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 27                 | 1                        | 0                     | 0                               | 1                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 28                 | 0                        | 0                     | 1                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 29                 | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 1                                        | 1                         | 0                     | 0                                 |
| 30                 | 0                        | 0                     | 1                               | 1                            | 0                   | 0                        | 0                                        | 1                         | 0                     | 0                                 |
| 31                 | 2                        | 0                     | 0                               | 0                            | 1                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 36                 | 1                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 40                 | 0                        | 0                     | 0                               | 1                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 41                 | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 1                     | 0                                 |
| 42                 | 0                        | 0                     | 1                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 45                 | 0                        | 0                     | 0                               | 0                            | 1                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |
| 111                | 1                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     | 0                                 |

Table 1b - Effective method line count distribution (normalized)

The table shown here only includes the first 40 rows which typically represents the most significant entries. Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName       | axon-messaging-5.0.2.jar | axon-common-5.0.2.jar | axon-server-connector-5.0.2.jar | axon-eventsourcing-5.0.2.jar | axon-test-5.0.2.jar | axon-modelling-5.0.2.jar | axon-spring-boot-autoconfigure-5.0.2.jar | axon-conversion-5.0.2.jar | axon-update-5.0.2.jar | axon-metric-5.0.2.jar |
|--------------------|--------------------------|-----------------------|---------------------------------|------------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|-----------------------|
| effectiveLineCount |                          |                       |                                 |                              |                     |                          |                                          |                           |                       |                       |
| 1                  | 54.369205                | 45.173745             | 41.56250                        | 50.413223                    | 49.156627           | 44.757033                | 58.012821                                | 55.494505                 | 57.142857             | 36.956                |
| 2                  | 13.908401                | 18.146718             | 22.34375                        | 12.396694                    | 16.626506           | 14.322251                | 25.320513                                | 10.989011                 | 6.493506              | 18.478                |
| 3                  | 11.660579                | 11.969112             | 12.03125                        | 10.082645                    | 7.951807            | 7.672634                 | 5.128205                                 | 6.593407                  | 3.896104              | 16.304                |
| 4                  | 5.310480                 | 5.019305              | 6.87500                         | 8.760331                     | 8.915663            | 9.207161                 | 1.923077                                 | 4.395604                  | 6.493506              | 6.521                 |
| 5                  | 4.298960                 | 4.633205              | 5.15625                         | 5.123967                     | 0.963855            | 5.882353                 | 2.884615                                 | 4.945055                  | 5.194805              | 2.173                 |
| 6                  | 2.584996                 | 4.633205              | 2.34375                         | 2.809917                     | 3.373494            | 4.603581                 | 2.243590                                 | 3.846154                  | 5.194805              | 4.347                 |
| 7                  | 1.882551                 | 1.801802              | 1.40625                         | 2.479339                     | 1.927711            | 2.301790                 | 1.602564                                 | 2.747253                  | 1.298701              | 4.347                 |
| 8                  | 1.994942                 | 2.187902              | 2.34375                         | 0.826446                     | 1.686747            | 2.301790                 | 0.641026                                 | 2.747253                  | 1.298701              | 5.434                 |
| 9                  | 1.011520                 | 1.415701              | 1.71875                         | 1.652893                     | 1.686747            | 3.324808                 | 0.320513                                 | 2.197802                  | 2.597403              | 1.086                 |
| 10                 | 0.814836                 | 0.772201              | 1.09375                         | 1.157025                     | 0.481928            | 0.767263                 | 0.320513                                 | 0.549451                  | 1.298701              | 0.000                 |
| 11                 | 0.561956                 | 0.900901              | 0.31250                         | 0.826446                     | 0.963855            | 0.511509                 | 0.000000                                 | 1.098901                  | 0.000000              | 0.000                 |
| 12                 | 0.280978                 | 0.900901              | 0.46875                         | 0.661157                     | 1.686747            | 1.023018                 | 0.320513                                 | 0.000000                  | 0.649351              | 0.000                 |
| 13                 | 0.224782                 | 0.643501              | 0.31250                         | 0.661157                     | 0.481928            | 1.278772                 | 0.320513                                 | 0.549451                  | 1.948052              | 3.260                 |
| 14                 | 0.112391                 | 0.643501              | 0.31250                         | 0.495868                     | 0.481928            | 0.255754                 | 0.000000                                 | 0.000000                  | 1.298701              | 0.000                 |
| 15                 | 0.168587                 | 0.128700              | 0.15625                         | 0.330579                     | 0.722892            | 0.255754                 | 0.000000                                 | 0.549451                  | 0.000000              | 1.086                 |
| 16                 | 0.196684                 | 0.257400              | 0.15625                         | 0.165289                     | 0.722892            | 0.255754                 | 0.000000                                 | 1.098901                  | 0.649351              | 0.000                 |
| 17                 | 0.112391                 | 0.386100              | 0.31250                         | 0.165289                     | 0.000000            | 0.511509                 | 0.320513                                 | 0.549451                  | 1.298701              | 0.000                 |
| 18                 | 0.028098                 | 0.128700              | 0.15625                         | 0.000000                     | 0.481928            | 0.511509                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.000                 |
| 19                 | 0.084293                 | 0.000000              | 0.00000                         | 0.165289                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.549451                  | 0.000000              | 0.000                 |
| 20                 | 0.056196                 | 0.000000              | 0.15625                         | 0.165289                     | 0.481928            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 21                 | 0.028098                 | 0.000000              | 0.31250                         | 0.165289                     | 0.240964            | 0.255754                 | 0.000000                                 | 0.000000                  | 1.298701              | 0.000                 |
| 22                 | 0.084293                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 23                 | 0.028098                 | 0.000000              | 0.00000                         | 0.000000                     | 0.240964            | 0.000000                 | 0.320513                                 | 0.000000                  | 0.000000              | 0.000                 |
| 24                 | 0.000000                 | 0.128700              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.000                 |
| 25                 | 0.056196                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 26                 | 0.000000                 | 0.128700              | 0.00000                         | 0.000000                     | 0.240964            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 27                 | 0.028098                 | 0.000000              | 0.00000                         | 0.165289                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 28                 | 0.000000                 | 0.000000              | 0.15625                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 29                 | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.320513                                 | 0.549451                  | 0.000000              | 0.000                 |
| 30                 | 0.000000                 | 0.000000              | 0.15625                         | 0.165289                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.549451                  | 0.000000              | 0.000                 |
| 31                 | 0.056196                 | 0.000000              | 0.00000                         | 0.000000                     | 0.240964            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 36                 | 0.028098                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 40                 | 0.000000                 | 0.000000              | 0.00000                         | 0.165289                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 41                 | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.000                 |
| 42                 | 0.000000                 | 0.000000              | 0.15625                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 45                 | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.240964            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |
| 111                | 0.028098                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.000                 |

Table 1b Chart 1 - Effective method line count distribution (normalized)

<Figure size 640x480 with 0 Axes>

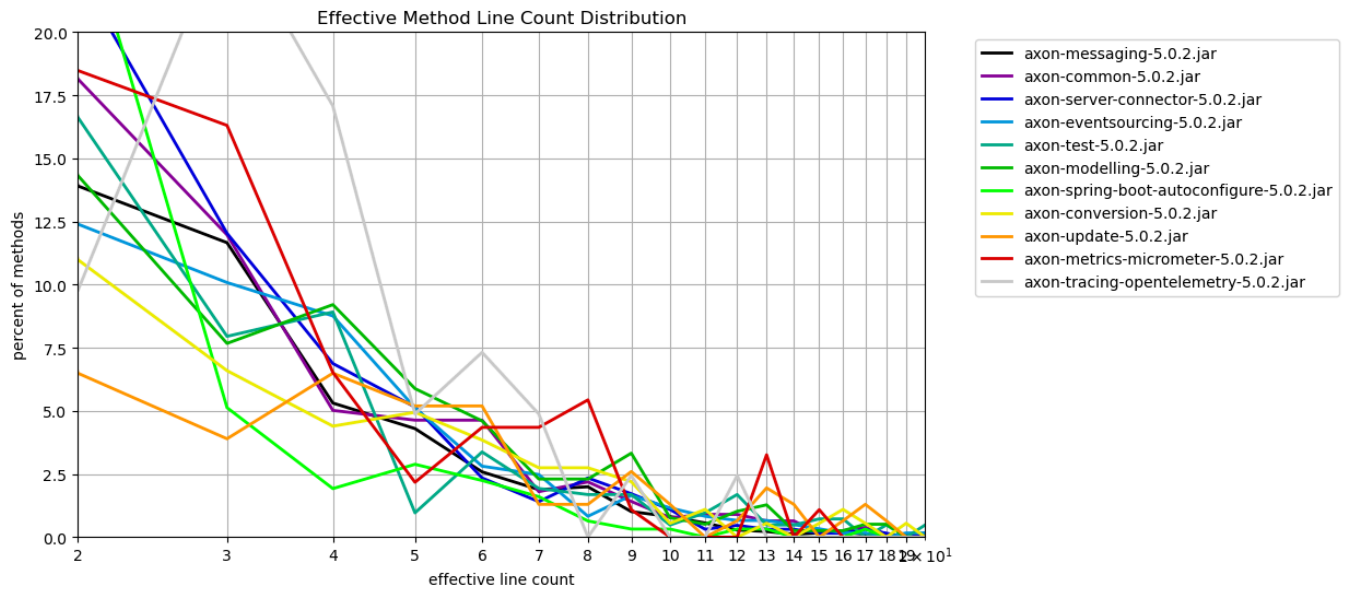


Table 1c - Top 30 packages with highest effective line counts

The following table shows the top 30 packages with the highest effective lines of code. The whole table can be found in the CSV report

`Effective_lines_of_method_code_per_package` .

|    | artifactName                         | fullNamePackage                                   | linesInPackage | methodCount | maxLinesMethod | maxLinesMethod                  |
|----|--------------------------------------|---------------------------------------------------|----------------|-------------|----------------|---------------------------------|
| 0  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 1417           | 397         | 111            |                                 |
| 1  | axon-messaging-5.0.2                 | org.axonframework.messaging.core                  | 1119           | 531         | 16             |                                 |
| 2  | axon-test-5.0.2                      | org.axonframework.test.fixture                    | 733            | 204         | 45             | ap                              |
| 3  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.annotation       | 702            | 235         | 23             |                                 |
| 4  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector            | 698            | 278         | 42             |                                 |
| 5  | axon-common-5.0.2                    | org.axonframework.common.configuration            | 668            | 262         | 26             | invo                            |
| 6  | axon-common-5.0.2                    | org.axonframework.common                          | 608            | 182         | 24             | getExactDirectSuperTypesOfParar |
| 7  | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore        | 599            | 233         | 19             |                                 |
| 8  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.event      | 540            | 162         | 21             | lambd                           |
| 9  | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore.jpa    | 457            | 141         | 20             |                                 |
| 10 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 455            | 130         | 25             |                                 |
| 11 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling         | 381            | 172         | 14             |                                 |
| 12 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork       | 339            | 168         | 21             |                                 |
| 13 | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.query      | 325            | 90          | 21             | co                              |
| 14 | axon-modelling-5.0.2                 | org.axonframework.modelling.entity.annotation     | 321            | 75          | 18             | createOptic                     |
| 15 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 320            | 121         | 11             |                                 |
| 16 | axon-test-5.0.2                      | org.axonframework.test.matchers                   | 318            | 101         | 21             |                                 |
| 17 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | 304            | 134         | 29             |                                 |
| 18 | axon-conversion-5.0.2                | org.axonframework.conversion.avro                 | 292            | 81          | 30             |                                 |
| 19 | axon-metrics-micrometer-5.0.2        | org.axonframework.extension.metrics.micrometer    | 281            | 87          | 15             | reg                             |
| 20 | axon-modelling-5.0.2                 | org.axonframework.modelling.entity                | 268            | 76          | 18             |                                 |
| 21 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot            | 266            | 147         | 9              |                                 |
| 22 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration    | 253            | 119         | 19             | i                               |
| 23 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 253            | 94          | 19             |                                 |
| 24 | axon-common-5.0.2                    | org.axonframework.common.caching                  | 234            | 85          | 13             | lambdacom                       |
| 25 | axon-common-5.0.2                    | org.axonframework.common.infra                    | 233            | 56          | 14             |                                 |
| 26 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 231            | 50          | 22             |                                 |
| 27 | axon-messaging-                      | org.axonframework.messaging.core.timeout          | 217            | 56          | 16             |                                 |

|    | artifactName         | fullPackageName                             | linesInPackage | methodCount | maxLinesMethod | maxL |
|----|----------------------|---------------------------------------------|----------------|-------------|----------------|------|
|    | 5.0.2                |                                             |                |             |                |      |
| 28 | axon-messaging-5.0.2 | org.axonframework.messaging.commandhandling | 216            | 100         | 18             |      |
| 29 | axon-messaging-5.0.2 | org.axonframework.messaging.eventhandling   | 211            | 86          | 11             |      |

## Table 1d - Top 30 methods with the highest effective line count

The following table shows the top 30 methods with the highest effective lines of code. The whole table can be found in the CSV report

Effective\_lines\_of\_method\_code\_per\_package .

| index | artifactName                            | fullPackageName                                   | maxLinesMethodType                                 |                           |
|-------|-----------------------------------------|---------------------------------------------------|----------------------------------------------------|---------------------------|
| 0     | 0 axon-messaging-5.0.2                  | org.axonframework.messaging.eventhandling.proc... | Coordinator\$CoordinationTask                      |                           |
| 1     | 2 axon-test-5.0.2                       | org.axonframework.test.fixture                    | Reporter                                           |                           |
| 2     | 4 axon-server-connector-5.0.2           | org.axonframework.axonserver.connector            | AxonServerConnectionManager\$Builder               |                           |
| 3     | 42 axon-update-5.0.2                    | org.axonframework.update.detection                | TestEnvironmentDetector                            |                           |
| 4     | 32 axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.annotation.ref... | AnnotationBasedEventSourcedEntityFactory           | ac                        |
| 5     | 61 axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.annotation        | AnnotationBasedEventCriteriaResolver\$WrappedEv... |                           |
| 6     | 18 axon-conversion-5.0.2                | org.axonframework.conversion.avro                 | AvroConverter                                      |                           |
| 7     | 17 axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | ConverterAutoConfiguration                         |                           |
| 8     | 65 axon-conversion-5.0.2                | org.axonframework.conversion.json                 | JacksonConverter                                   |                           |
| 9     | 5 axon-common-5.0.2                     | org.axonframework.common.configuration            | DefaultAxonApplication\$AxonConfigurationImpl      |                           |
| 10    | 10 axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | JdbcTokenStore                                     |                           |
| 11    | 6 axon-common-5.0.2                     | org.axonframework.common                          | TypeReflectionUtils                                | getExactDirectSuperTypesC |
| 12    | 3 axon-messaging-5.0.2                  | org.axonframework.messaging.core.annotation       | MethodInvokingMessageHandlingMember                |                           |
| 13    | 84 axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.util       | AbstractQualifiedBeanCondition                     |                           |
| 14    | 26 axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | JpaTokenStore                                      |                           |
| 15    | 40 axon-update-5.0.2                    | org.axonframework.update                          | UpdateCheckerHttpClient                            |                           |
| 16    | 54 axon-modelling-5.0.2                 | org.axonframework.modelling.annotation            | AnnotationBasedEntityEvolvingComponent             |                           |
| 17    | 46 axon-update-5.0.2                    | org.axonframework.update.api                      | UpdateCheckResponse                                |                           |
| 18    | 16 axon-test-5.0.2                      | org.axonframework.test.matchers                   | DeepEqualsMatcher                                  |                           |
| 19    | 13 axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.query      | QueryConverter                                     |                           |
| 20    | 8 axon-server-connector-5.0.2           | org.axonframework.axonserver.connector.event      | AggregateBasedAxonServerEventStorageEngine         | I                         |
| 21    | 12 axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork       | UnitOfWork\$UnitOfWorkProcessingContext            |                           |
| 22    | 34 axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.command    | CommandConverter                                   |                           |
| 23    | 31 axon-test-5.0.2                      | org.axonframework.test.server                     | AxonServerContainerUtils                           |                           |
| 24    | 9 axon-eventsourcing-5.0.2              | org.axonframework.eventsourcing.eventstore.jpa    | GapAwareTrackingTokenOperations                    |                           |
| 25    | 7 axon-eventsourcing-5.0.2              | org.axonframework.eventsourcing.eventstore        | AnnotationBasedTagResolver                         |                           |
| 26    | 22 axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration    | MessagingConfigurationDefaults                     |                           |
| 27    | 23 axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | TrackerStatus                                      |                           |

|    | index | artifactName         | fullPackageName                               | maxLinesMethodType       |       |
|----|-------|----------------------|-----------------------------------------------|--------------------------|-------|
| 28 | 14    | axon-modelling-5.0.2 | org.axonframework.modelling.entity.annotation | AnnotatedEntityMetamodel | creat |
| 29 | 20    | axon-modelling-5.0.2 | org.axonframework.modelling.entity            | ConcreteEntityMetamodel  |       |

## Cyclomatic Complexity

Table 2a - Cyclomatic method complexity distribution

This table shows the distribution of the cyclomatic complexity of methods per artifact. For each artifact the number of methods with the cyclomatic complexity = 1,2,3,... is shown to get an overview of how cyclomatic complexity is distributed over methods.

Only the 15 artifacts with the highest method count sum and their cyclomatic method complexity distribution (limited by 40) is shown here. The whole table can be found in the CSV report [Cyclomatic\\_Method\\_Complexity\\_Distribution](#).

Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName         | axon-messaging-5.0.2.jar | axon-common-5.0.2.jar | axon-server-connector-5.0.2.jar | axon-eventsourcing-5.0.2.jar | axon-test-5.0.2.jar | axon-modelling-5.0.2.jar | axon-spring-boot-autoconfigure-5.0.2.jar | axon-conversion-5.0.2.jar | axon-update-5.0.2.jar | ax metr microme 5.0.2 |
|----------------------|--------------------------|-----------------------|---------------------------------|------------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|-----------------------|
| cyclomaticComplexity |                          |                       |                                 |                              |                     |                          |                                          |                           |                       |                       |
| 1                    | 2926                     | 544                   | 536                             | 463                          | 314                 | 311                      | 288                                      | 137                       | 119                   |                       |
| 2                    | 320                      | 91                    | 57                              | 59                           | 40                  | 27                       | 11                                       | 14                        | 16                    |                       |
| 3                    | 153                      | 74                    | 26                              | 38                           | 33                  | 23                       | 7                                        | 8                         | 5                     |                       |
| 4                    | 83                       | 31                    | 7                               | 20                           | 7                   | 16                       | 2                                        | 11                        | 1                     |                       |
| 5                    | 29                       | 10                    | 5                               | 8                            | 9                   | 7                        | 1                                        | 2                         | 3                     |                       |
| 6                    | 32                       | 7                     | 5                               | 6                            | 5                   | 5                        | 0                                        | 3                         | 4                     |                       |
| 7                    | 8                        | 6                     | 1                               | 5                            | 2                   | 1                        | 1                                        | 2                         | 2                     |                       |
| 8                    | 3                        | 8                     | 0                               | 4                            | 2                   | 1                        | 0                                        | 1                         | 0                     |                       |
| 9                    | 1                        | 3                     | 2                               | 0                            | 1                   | 0                        | 2                                        | 1                         | 1                     |                       |
| 10                   | 0                        | 1                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 1                         | 0                     |                       |
| 11                   | 1                        | 1                     | 0                               | 1                            | 1                   | 0                        | 0                                        | 0                         | 1                     |                       |
| 12                   | 0                        | 1                     | 1                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     |                       |
| 13                   | 1                        | 0                     | 0                               | 1                            | 1                   | 0                        | 0                                        | 0                         | 0                     |                       |
| 17                   | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 1                         | 1                     |                       |
| 21                   | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 1                         | 0                     |                       |
| 25                   | 1                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     |                       |
| 41                   | 0                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 1                     |                       |
| 58                   | 1                        | 0                     | 0                               | 0                            | 0                   | 0                        | 0                                        | 0                         | 0                     |                       |

Table 2b - Cyclomatic method complexity distribution (normalized)

The table shown here only includes the first 40 rows which typically represents the most significant entries. Have a look below to find out which packages and methods have the



highest effective lines of code.

|                      | artifactName | axon-messaging-5.0.2.jar | axon-common-5.0.2.jar | axon-server-connector-5.0.2.jar | axon-eventsourcing-5.0.2.jar | axon-test-5.0.2.jar | axon-modelling-5.0.2.jar | axon-spring-boot-autoconfigure-5.0.2.jar | axon-conversion-5.0.2.jar | axon-update-5.0.2.jar | axon-metrics-micrometer-5.0.2.jar |
|----------------------|--------------|--------------------------|-----------------------|---------------------------------|------------------------------|---------------------|--------------------------|------------------------------------------|---------------------------|-----------------------|-----------------------------------|
| cyclomaticComplexity |              |                          |                       |                                 |                              |                     |                          |                                          |                           |                       |                                   |
| 1                    |              | 82.214105                | 70.012870             | 83.75000                        | 76.528926                    | 75.662651           | 79.539642                | 92.307692                                | 75.274725                 | 77.272727             | 93.4                              |
| 2                    |              | 8.991290                 | 11.711712             | 8.90625                         | 9.752066                     | 9.638554            | 6.905371                 | 3.525641                                 | 7.692308                  | 10.389610             | 3.2                               |
| 3                    |              | 4.298960                 | 9.523810              | 4.06250                         | 6.280992                     | 7.951807            | 5.882353                 | 2.243590                                 | 4.395604                  | 3.246753              | 1.0                               |
| 4                    |              | 2.332116                 | 3.989704              | 1.09375                         | 3.305785                     | 1.686747            | 4.092072                 | 0.641026                                 | 6.043956                  | 0.649351              | 0.0                               |
| 5                    |              | 0.814836                 | 1.287001              | 0.78125                         | 1.322314                     | 2.168675            | 1.790281                 | 0.320513                                 | 1.098901                  | 1.948052              | 0.0                               |
| 6                    |              | 0.899129                 | 0.900901              | 0.78125                         | 0.991736                     | 1.204819            | 1.278772                 | 0.000000                                 | 1.648352                  | 2.597403              | 2.0                               |
| 7                    |              | 0.224782                 | 0.772201              | 0.15625                         | 0.826446                     | 0.481928            | 0.255754                 | 0.320513                                 | 1.098901                  | 1.298701              | 0.0                               |
| 8                    |              | 0.084293                 | 1.029601              | 0.00000                         | 0.661157                     | 0.481928            | 0.255754                 | 0.000000                                 | 0.549451                  | 0.000000              | 0.0                               |
| 9                    |              | 0.028098                 | 0.386100              | 0.31250                         | 0.000000                     | 0.240964            | 0.000000                 | 0.641026                                 | 0.549451                  | 0.649351              | 0.0                               |
| 10                   |              | 0.000000                 | 0.128700              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.549451                  | 0.000000              | 0.0                               |
| 11                   |              | 0.028098                 | 0.128700              | 0.00000                         | 0.165289                     | 0.240964            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.0                               |
| 12                   |              | 0.000000                 | 0.128700              | 0.15625                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |
| 13                   |              | 0.028098                 | 0.000000              | 0.00000                         | 0.165289                     | 0.240964            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |
| 17                   |              | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.549451                  | 0.649351              | 0.0                               |
| 21                   |              | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.549451                  | 0.000000              | 0.0                               |
| 25                   |              | 0.028098                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |
| 41                   |              | 0.000000                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.649351              | 0.0                               |
| 58                   |              | 0.028098                 | 0.000000              | 0.00000                         | 0.000000                     | 0.000000            | 0.000000                 | 0.000000                                 | 0.000000                  | 0.000000              | 0.0                               |

Table 2b Chart 1 - Cyclomatic method complexity distribution (normalized)

<Figure size 640x480 with 0 Axes>

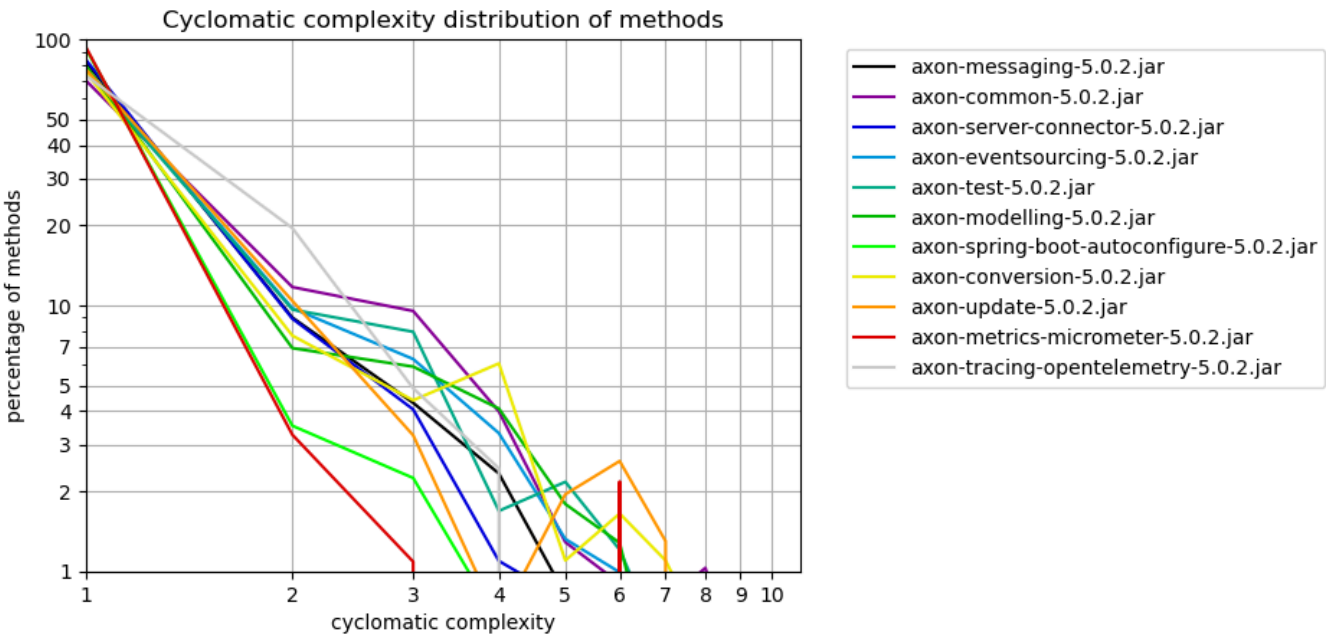


Table 2c - Top 30 packages with highest cyclomatic complexity

The following table shows the top 30 packages with the highest cyclomatic complexity. The whole table can be found in the CSV report

Effective\_lines\_of\_method\_code\_per\_package .

|    | artifactName                         | fullPackageName                                   | complexityInPackage | methodCount | maxComplexity | maxComplexity             |
|----|--------------------------------------|---------------------------------------------------|---------------------|-------------|---------------|---------------------------|
| 1  | axon-messaging-5.0.2                 | org.axonframework.messaging.core                  | 731                 | 531         | 9             |                           |
| 0  | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 546                 | 397         | 25            |                           |
| 6  | axon-common-5.0.2                    | org.axonframework.common                          | 383                 | 182         | 9             | doConstructionWithOptiona |
| 7  | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore        | 375                 | 233         | 8             |                           |
| 3  | axon-messaging-5.0.2                 | org.axonframework.messaging.core.annotation       | 371                 | 235         | 13            | resolve                   |
| 5  | axon-common-5.0.2                    | org.axonframework.common.configuration            | 358                 | 262         | 8             | hasC                      |
| 4  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector            | 357                 | 278         | 12            |                           |
| 2  | axon-test-5.0.2                      | org.axonframework.test.fixture                    | 322                 | 204         | 11            | appendEven                |
| 15 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 249                 | 121         | 7             |                           |
| 12 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.unitofwork       | 224                 | 168         | 6             |                           |
| 8  | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.event      | 218                 | 162         | 4             | spl                       |
| 9  | axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.eventstore.jp...  | 217                 | 141         | 8             | loadKeyViolate            |
| 11 | axon-messaging-5.0.2                 | org.axonframework.messaging.queryhandling         | 216                 | 172         | 4             | subscribe                 |
| 10 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 196                 | 130         | 8             | up                        |
| 17 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | 177                 | 134         | 9             | registerHandlerIn         |
| 16 | axon-test-5.0.2                      | org.axonframework.test.matchers                   | 173                 | 101         | 8             | m                         |
| 18 | axon-conversion-5.0.2                | org.axonframework.conversion.avro                 | 159                 | 81          | 21            |                           |
| 21 | axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot            | 150                 | 147         | 2             | th                        |
| 24 | axon-common-5.0.2                    | org.axonframework.common.caching                  | 147                 | 85          | 8             |                           |
| 23 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | 147                 | 94          | 8             |                           |
| 37 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.anno... | 143                 | 60          | 58            | parse                     |
| 22 | axon-messaging-5.0.2                 | org.axonframework.messaging.core.configuration    | 131                 | 119         | 4             | lambda <i>registerDec</i> |
| 14 | axon-modelling-5.0.2                 | org.axonframework.modelling.entity.annotation     | 130                 | 75          | 6             | createCommandTarg         |
| 13 | axon-server-connector-5.0.2          | org.axonframework.axonserver.connector.query      | 127                 | 90          | 6             | responseSe                |
| 20 | axon-modelling-5.0.2                 | org.axonframework.modelling.entity                | 119                 | 76          | 7             | hand                      |
| 43 | axon-messaging-5.0.2                 | org.axonframework.messaging.eventstreaming        | 119                 | 92          | 4             |                           |
| 28 | axon-messaging-5.0.2                 | org.axonframework.messaging.commandhandling       | 117                 | 100         | 4             |                           |

|    | artifactName                  | fullPackageName                                   | complexityInPackage | methodCount | maxComplexity | maxComplexity          |
|----|-------------------------------|---------------------------------------------------|---------------------|-------------|---------------|------------------------|
| 29 | axon-messaging-5.0.2          | org.axonframework.messaging.eventhandling         | 106                 | 86          | 4             | an                     |
| 19 | axon-metrics-micrometer-5.0.2 | org.axonframework.extension.metrics.micrometer    | 99                  | 87          | 6             | registerComponentWithD |
| 26 | axon-messaging-5.0.2          | org.axonframework.messaging.eventhandling.proc... | 99                  | 50          | 6             |                        |

## Table 2d - Top 30 methods with highest cyclomatic complexity

The following table shows the top 30 packages containing the methods with the highest cyclomatic complexity. The whole table can be found in the CSV report

[Effective\\_lines\\_of\\_method\\_code\\_per\\_package](#) .

| index | artifactName                            | fullPackageName                                   | maxComplexityType                                  | maxComplexity            |
|-------|-----------------------------------------|---------------------------------------------------|----------------------------------------------------|--------------------------|
| 0     | 37 axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.anno... | MethodSequencingPolicyEventHandlerDefinition\$S... | parse                    |
| 1     | 42 axon-update-5.0.2                    | org.axonframework.update.detection                | TestEnvironmentDetector                            |                          |
| 2     | 0 axon-messaging-5.0.2                  | org.axonframework.messaging.eventhandling.proc... | Coordinator\$CoordinationTask                      |                          |
| 3     | 18 axon-conversion-5.0.2                | org.axonframework.conversion.avro                 | AvroConverter                                      |                          |
| 4     | 65 axon-conversion-5.0.2                | org.axonframework.conversion.json                 | JacksonConverter                                   |                          |
| 5     | 46 axon-update-5.0.2                    | org.axonframework.update.api                      | UpdateCheckResponse                                | fr                       |
| 6     | 32 axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.annotation.ref... | AnnotationBasedEventSourcedEntityFactory           | addEntityCreator         |
| 7     | 31 axon-test-5.0.2                      | org.axonframework.test.server                     | AxonServerContainer                                |                          |
| 8     | 3 axon-messaging-5.0.2                  | org.axonframework.messaging.core.annotation       | MessageStreamResolverUtils                         | resolv                   |
| 9     | 4 axon-server-connector-5.0.2           | org.axonframework.axonserver.connector            | AxonServerConnectionManager\$Builder               |                          |
| 10    | 49 axon-common-5.0.2                    | org.axonframework.common.jdbc                     | ConnectionWrapperFactory                           | lam                      |
| 11    | 61 axon-eventsourcing-5.0.2             | org.axonframework.eventsourcing.annotation        | AnnotationBasedEventCriteriaResolver\$WrappedEv... |                          |
| 12    | 2 axon-test-5.0.2                       | org.axonframework.test.fixture                    | Reporter                                           | appendEve                |
| 13    | 53 axon-common-5.0.2                    | org.axonframework.common.lock                     | PessimisticLockFactory\$DisposableLock             |                          |
| 14    | 6 axon-common-5.0.2                     | org.axonframework.common                          | ConstructorUtils                                   | doConstructionWithOption |
| 15    | 1 axon-messaging-5.0.2                  | org.axonframework.messaging.core                  | MessageStreamUtils\$Reducer                        |                          |
| 16    | 17 axon-spring-boot-autoconfigure-5.0.2 | org.axonframework.extension.springboot.autoconfig | InterceptorAutoConfiguration                       | registerHandlerIn        |
| 17    | 9 axon-eventsourcing-5.0.2              | org.axonframework.eventsourcing.eventstore.jp     | SQLErrorCodesResolver                              | loadKeyViol              |
| 18    | 24 axon-common-5.0.2                    | org.axonframework.common.caching                  | EhCacheAdapter\$CacheEventListenerAdapter          |                          |
| 19    | 23 axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | EventTrackerStatus                                 |                          |
| 20    | 5 axon-common-5.0.2                     | org.axonframework.common.configuration            | DefaultComponentRegistry                           | has                      |
| 21    | 68 axon-common-5.0.2                    | org.axonframework.common.annotation               | AnnotationUtils                                    | collec                   |
| 22    | 7 axon-eventsourcing-5.0.2              | org.axonframework.eventsourcing.eventstore        | AggregateBasedConsistencyMarker                    |                          |
| 23    | 16 axon-test-5.0.2                      | org.axonframework.test.matchers                   | SequenceMatcher                                    | r                        |
| 24    | 10 axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | JdbcTokenStore                                     | u                        |
| 25    | 36 axon-modelling-5.0.2                 | org.axonframework.modelling.entity.child          | FieldChildEntityFieldDefinition                    | evolveParentBasedO       |
| 26    | 20 axon-modelling-5.0.2                 | org.axonframework.modelling.entity                | ConcreteEntityMetamodel                            | han                      |
| 27    | 25 axon-common-5.0.2                    | org.axonframework.common.infra                    | FilesystemStyleComponentDescriptor\$TreeRenderer   | renc                     |
| 28    | 72 axon-common-5.0.2                    | org.axonframework.common.util                     | PriorityRunnable                                   |                          |
| 29    | 15 axon-messaging-5.0.2                 | org.axonframework.messaging.eventhandling.proc... | ReplayToken                                        |                          |

| index | artifactName | fullPackageName | maxComplexityType | maxComplex |
|-------|--------------|-----------------|-------------------|------------|
|       | 5.0.2        |                 |                   |            |